

ITE 387 ADVANCED PROGRAMMING

FINAL PROJECT PROPOSAL

Leader/Project Manager:

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Team Members

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Project Title

AnchorWatch: Data Science for Stablecoin Stability and Macroeconomic Integration

Project Overview

Summary:

AnchorWatch is an AI-enhanced financial intelligence platform designed to track and analyze stablecoin flows alongside global macroeconomic indicators (e.g., GDP, inflation, interest rates). It enables users to explore real-time data visualizations, identify abnormal trends, and gain insights into the interplay between blockchain-based assets and economic health. The platform is delivered as a web application for administrative tasks (using React + TypeScript) and a mobile application for users (using Flutter for Android/iOS). The web app serves administrators for user management, while the mobile app provides an interface for user customization. Built with modern tools and cloud infrastructure, AnchorWatch prioritizes accessibility and scalability.

Objectives

- Monitor and visualize global stablecoin flows using blockchain data.
- Integrate macroeconomic indicators to contextualize crypto-economic trends.
- Detects volatility patterns and mint/burn anomalies in stablecoin usage.
- Summarize market and economic movements using LLM-generated insights.
- Provide interactive dashboards via a mobile application for users.

- Identify regional or behavioral usage patterns using data clustering.
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Problem Statement

Stablecoins are critical to global digital finance, yet existing monitoring tools are fragmented, overly technical, or lack economic context. Macroeconomic indicators like inflation or GDP are rarely visualized alongside on-chain data, hindering risk assessment, market shift detection, and informed decision-making for developers, analysts, and institutions.

Proposed Solution

AnchorWatch unifies blockchain analytics, macroeconomic data, and AI into a streamlined platform. It collects on-chain data (via Etherscan APIs and Coingecko) and off-chain economic data (via the World Bank, IMF, and FRED), processes them through ETL pipelines, and presents them in intuitive visual dashboards. LLMs provide intelligent summarization and automated anomaly detection, enhancing the speed and accuracy of financial insights. A mobile interface that displays dashboards, while an admin panel manages users.

Key Features:

- **Mobile Dashboards (User-Facing):** Display stablecoin supply, mint/burn events, and more; compare blockchain activity to macroeconomic trends (e.g, GDP, inflation, interest rates).
 - **Heatmaps and Time-Series (Mobile):** Visualize global stablecoin movements and trends.
 - **LLM-Generated Summaries (Mobile):** Provide textual insights into significant market changes.
 - **Mobile App (Flutter):** Interface for users, mostly the analytics part
 - **Admin Web Application:** Manage user accounts, configure ETL/LLM pipelines.
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Scope

Web Application (React + TypeScript) - Admin

- **Admin Panel:** Interface for managing user accounts, configuring ETL schedules, and tuning LLM prompts.
- **User Management:** View user list and update profiles, stored in MongoDB.
- **System Configuration:** Schedule Celery tasks, manage LLM prompts (LangChain), and set anomaly detection rules.
- **Authentication:** Admin login via Django with secure credentials management.

Mobile Application (Flutter) - User

- **Condensed Dashboard:** Live wallet/stablecoin data and macroeconomic trends (GDP, inflation).
- **Visual Heatmap:** Regional stablecoin activity visualization.
- **Minimalist UI:** Streamlined for browsing summaries and charts on Android/iOS.
- **LLM Summaries:** AI-generated insights filterable by timeframe or asset.

Backend / ETL / LLM Pipeline

- **Framework:** Django REST Framework for APIs serving both the admin web app and the user mobile app.
- **Task Queues:** Celery + Redis for scheduling ETL tasks (blockchain/macro data) and LLM summarization.

- **NLP Engine:** LangChain, still looking for models.
- **Admin APIs:** Support user management, ETL configuration.

Database / Storage

- **MongoDB Atlas:** Centralized, schema-flexible storage for event logs, anomaly flags, user sessions, LLM outputs, and admin configurations.
- **User Data:** Stores profiles

Cloud Infrastructure / DevOps

- **Cloud Provider:** AWS (student-tier EC2, S3, RDS as needed).
 - **Containerization:** Docker for consistent deployment.
 - **Secrets Management:** AWS Secrets Manager for API keys and credentials.
 - **Infrastructure as Code:** Terraform for automated deployment.
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Use Cases

- **Financial Analysts:** Monitor stablecoin behavior alongside inflation/GDP shifts via mobile app for risk assessment.
- **Developers:** Explore macro-on-chain relationships via APIs or mobile dashboards.
- **Users:** Check for mint anomalies or rapid depegging to inform strategies. Access AI-generated summaries for macro movements.
- **Administrators:** Manage user accounts, configure pipelines.